Input Offset Cancellation Trimming Technique for Operational Amplifiers

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***Abstract*— This paper presents a novel input offset cancellation technique dedicated for analog building blocks. This technique provides an injected current to the analog input signal that is a function of the offset. The proposed technique provides a suitable method to the post-package offset cancellation. Unlike, other convention techniques, the proposed technique avoids the potentially stability issue, consumes a low silicon area, low power consumption, and temperature independent. The input offset cancellation circuit is attached to the input stage of the differential input single output (DISO) amplifier to revoke undesired DC, offset voltage ($V_{os,\ out}$), based on trimming current (I_{Trim}). DISO amplifier with input offset cancellation circuit is simulated and comprehend a DC gain of 78 dB, a unity-gain frequency (GBW) of 58 MHz associated with a phase margin of 67°. Moreover, the total consumption power is 0.72 mW. The statistical analysis with practical different coefficients provided from foundry demonstrates offset voltage less than 200 μv. The circuit is simulated in IBM 0.13µm CMOS technology with a single power supply 1.5-V.**

Keywords- Analog circuit; Offset cancellation; Post-packging; Trimming current; DISO.

I. INTRODUCTION

Steady advances in design of MOS integrated circuit increases the direction for realization analog and digital circuits on a single chip. Frequently, the quality of the analog portion of such chips is limited by the offset voltage ($V_{os,\ out}$) exhibited by integrated differential amplifiers. Differential amplifier is used in many vital applications whether that was analog or digital applications such as, an analog circuit of an orthogonal frequency divisions multiplexing (OFDM) wireless communication system [1], optical communication system, and audio/video processing system [2,3]. So, it is mandatory to identify the dominant reasons of $V_{os,\ out}$ to find a qualified settlement to revoke.

The offset voltage is generated from transistor mismatches due to technology scaling trend and reduction in the feature size. Furthermore, the mechanical stress [4] is an additional factor to cause a DC error or $V_{os,\ out}$. Usually, the offset voltage is specified by assuming a single source of error at the input terminals as shown in Fig. 1(a). The value of this imaginary voltage source represents the input referred offset voltage ($V_{os,\ in}$) of the amplifier [5].

The harmful effects of DC error lies in the limitation of the amplifier ability to resolve a small DC input voltage smaller than the $V_{os,\ in}$. This leads to reduce the output swing and drives the latter stages into nonlinear operation [6]. Therefore, DC offset cancellation is imperative way to reduce the inherent offset as shown in Fig. 1(b).

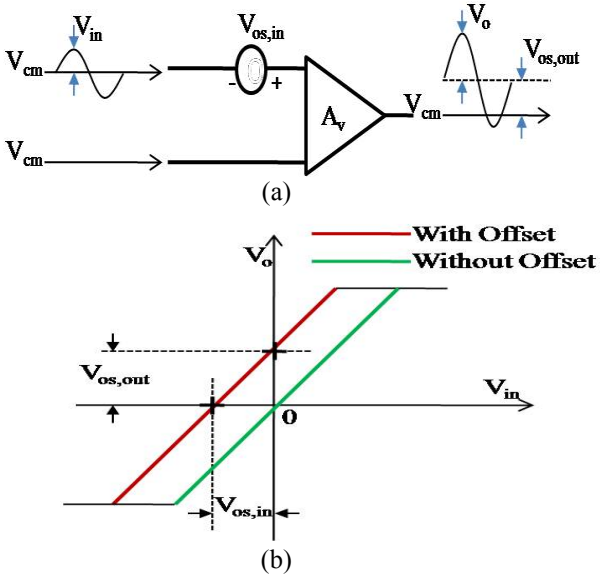


Fig. 1. (a) The input referred offset voltage ($V_{os,\ in}$) of the DISO amplifier representation. (b) DC characteristic curve of the DISO amplifier.

Many techniques have been reported [7, 8, 9] and used in the past to cancel out the offset voltage. Based on the trimming during the production process, all trimming techniques can be classified into two categories: pre-package trim and post-package trim. The pre-package trim includes all trim techniques that take place before the silicon die is packaged into its final package. On the other hand, post-package trim is done after packaging the part. The pre-package trim is a critical one, since the packaging process can lead to significant offset shifts of the part. Especially, the majority of all parts are shipped by plastic packages which inflict high mechanical stresses to silicon die [4]. These mentioned mechanical stresses are converted into offset voltage due to the piezoelectric prosperities of silicon. These offset voltage changes randomly after packaging.

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The most common method for pre-package compensation of the aforementioned offset error is to determine the differential input voltage offset error. Then, add or subtract such voltage offset error from one of the differential input terminals. For example, Ac coupling technique [5] in which a capacitor is used in the signal's path at input or output. The penalty of this method is signal loss and large area due to larger capacitor. Another method is the feedback offset cancellation [10, 11] technique which tends to extract the amplified offset from the output and feedback to the input to eliminate any potential DC offset. This method suffers from loop delay, loop gain, and potential stability issues. Finally, feedforward offset cancellation technique [12] which uses high consumed area and high power consumption. Despite the strenuous efforts to remove offset voltage before packaging, but it will most likely show a non-zero offset after being assembled [4]. Therefore, pre-package trim is not able to guarantee very high accuracies of the trimmed offset voltage.

So, post-package trim technique is a mandatory method because it takes into account offset shift due to package. Therefore, very high trim accuracies can be achieved. In the former arts of post-package trim techniques make permanent changes to an IC in order to store the offset compensation voltage. For example, include trimming resistors with a laser, blowing links with a laser, shorting out zener diodes by a high reverse voltage, lowing metal fuses with a high current, modifying the resistivity of a poly-silicon resistor with a high current, and storing digital bits onto a non-volatile memory (e.g. flash) [7]. In this work, a novel input offset cancellation technique for DISO amplifier is discussed to overcome the above mentioned issues. The proposed circuit is based on injected current to the analog input signal in which an additional DC voltage is generated to revoke the offset.

This article is organized as following: The proposed circuit is described in section II. In section III, simulation results are presented. Finally, conclusion is presented in section IV.

II. THE PROPOSED CIRCUIT

The block diagram shown in Fig. 2 is an exemplary embodiment to the proposed settlement for canceling the DC error. The proposed solution suggests an offset cancellation circuit controlled by injected current which is connected to one of amplifier's terminals. The offset cancellation circuit can recover the $V_{os,\ in}$ by adding a DC voltage. This diagram serves as a starting point for the following discussion. Each block will be discussed in detail for the specific implementation.

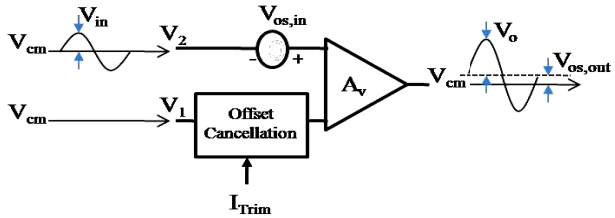


Fig. 2. The block diagram of the proposed solution.

A. Design a CMOS Differential Amplifier

CMOS differential amplifier shown in Fig. 3 consists of PMOS-differential pair M_1 and M_2, as the first stage, with NMOS active load M_3 and M_4 biased by PMOS transistor M_7. PMOS-differential pair is followed by the second stage, common source amplifier. The second stage consists of NMOS transistor M_5 with PMOS active load M_6. Both two stage are biased by transferring the reference current (I_{ref}) through M_8. Dimension of M_{13}, diode connection, is set according to available I_{ref}. Stability issues can be handled by using compensation network composed from compensation capacitor (C_c) and nulling resistor (R_z) [6]. The two-stage CMOS differential amplifier with compensation network shown in Fig. 3(a, b), is analyzed to be a part of the proposed solution.

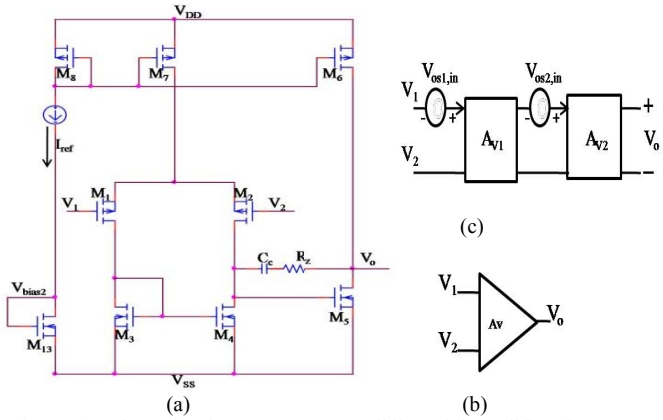


Fig. 3. (a) Schematic of CMOS two-stage differential amplifier. (b) Its symbol.(c) Input referred offset voltage for each stage representation.

There are several methodologies mentioned in [5, 6, 13, 14] to design a CMOS two stage differential amplifier. One of the most recent one is g_m/I_D methodology, described in [14]. These methodology has a lot of benefits as strongly related to the performances of analog circuits. The methodology design of two stage differential amplifier described in [6, 14] is used to evaluate the dimensions of all transistors.

$$V_{os,in} = V_{os1,in} + \frac{V_{os2,in}}{A_{v1}} \quad (1)$$

$$V_{os,in} = \left\{ \sqrt{\frac{2I_{D3}}{\beta_{n3}}} \left[\frac{\Delta\left(\frac{W}{L}\right)}{2\left(\frac{W}{L}\right)} \right]_3 + \Delta V_{THn} \right\} \frac{g_{m3}}{g_{m1}} + \sqrt{\frac{2I_{D1}}{\beta_{p1}}} \left[\frac{\Delta\left(\frac{W}{L}\right)}{2\left(\frac{W}{L}\right)} \right]_1 + \Delta V_{THp} + \frac{V_{os2,in}}{A_{v1}} \quad (2)$$

Where;

$V_{os,in}$: is the input referred offset voltage of DISO amplifier.
$V_{os1,in}$: is the input referred offset voltage of 1st stage of DISO amplifier.
$V_{os2,in}$: is the input referred offset voltage of 2nd stage of DISO amplifier.
A_{v1} : is the voltage gain of 1st stage.
$I_{D1(3)}$: is the DC current of $M_{1(3)}$.
β_{p1} : is the gain process factor of PMOS, M_1.
β_{n3} : is the gain process factor of NMOS, M_3.
$\frac{W}{L}$: is the width to length ratio of MOST.
ΔV_{THn} : is the difference between threshold of M_3 and M_4.
ΔV_{THp} : is the difference between threshold of M_1 and M_2.
$g_{m1(3)}$: is the trans-conductance of $M_{1(3)}$.

In addition to used methodology [6 ,14], it is necessary to keep in mind the design and layout considerations to reduce any systematic and DC error [16]. Moreover, the first stage has superior priority and has a dominant effect on the offset voltage during the design with respect to design the second stage. So, $V_{os1,\ in}$ represented in Fig. 3(c) should be reduced according to (1).

B. Design the Offset Cancellation Circuit

Based on the prior art in [17], the offset cancellation technique circuit is mainly a two cascaded DC level shifters, common drain amplifiers. The schematic circuit and its symbol are presented in Fig. 4(a, b). The first DC level shift composes of PMOS M_9 with PMOS active load M_{10}. And, the other consists of NMOS M_{11} with NMOS active load M_{12}.

M_{10} is fed from PMOS diode connection M_{14}, forced in saturation mode. I_{Trim} is transferred from M_{14} to M_{10} by the same factor for equal dimensions. I_{Trim} is varied according to value of the potentiometer (POT) as shown in Fig. 4(c). Both M_{10} and M_{14} perform a current mirror. Also, M_{12} is bias from NMOS M_{13}, diode connection, shown in Fig. 3 (a).

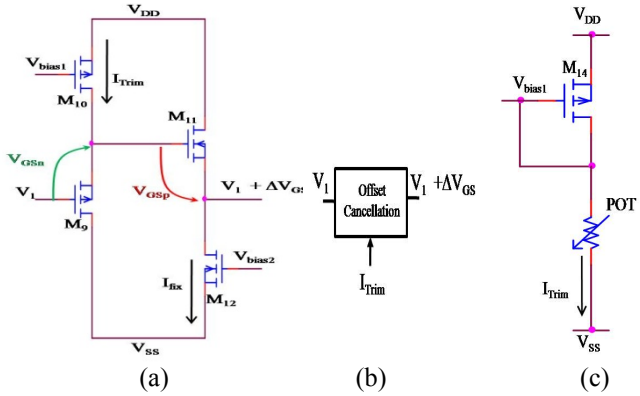


Fig. 4. (a) Schematic of the offset cancellation circuit. (b) its symbol. (c) Bias circuit with POT.

The proposed cell has a main function is to generate an additional offset voltage, ΔV_{GS} which is the difference between V_{GS} of M_9 and M_{11}, to cancel out the original offset generated from mismatches and mechanical stresses. This cell is connected to one of the differential pairs of DISO amplifier to compensate any DC error at the output. Using the proposed cell, an analog signal control line is provided to DISO amplifier.

$$V_{OS,\ in} = V_{OS,in}\Big|_{\substack{\text{due to mismatches},\\ \text{mechanical stresses}}} + \Delta V_{GS} \tag{3}$$

$$\Delta V_{GS} = \sqrt{\frac{2I_{Trim}}{\beta_{p9}}} - \sqrt{\frac{2I_{fix}}{\beta_{n11}}} + \Delta V_{TH} \tag{4}$$

$$I_{D14} = \frac{\beta_{p14}}{2}(V_{SG14} - V_{TH14})^2 \tag{5}$$

$$I_{Trim} = \frac{V_{DD} - V_{SG14}}{R} \tag{6}$$

Where;

$\beta_{p9(14)}$: is the gain process factor of PMOS , $M_{9(14)}$.
β_{n11} : is the gain process factor of NMOS, M_{11}.
I_{fix} : is the DC current of M_{12}.
ΔV_{TH} : is the difference between the threshold of M_9 and M_{11}.
V_{SG14} : is the DC voltage between the source and the gate of M_{14}.
V_{TH14} : is the threshold of the PMOS, M_{14}.
R : is the value of potentiometer resistor, POT.
V_{DD} : is the positive power supply.

The relation between $V_{OS,\ in}$ and I_{Trim} is expressed by (3) and (4). These relations indicate that, the designer is free to control the offset voltage after packaging the silicon die. It is clear form (4), that ΔV_{GS} is independent on temperature if threshold of NMOS and PMOS are equal. ΔV_{GS} is mainly function of I_{Trim} so, external analog control line is provided. Using (5) and (6), potentiometer enables the customer to control on I_{Trim} and therefore control the offset voltage.

$$\frac{V_o}{V_i} = \frac{g_{m9}.R_{10}}{1+g_{m9}.R_{10}} \cdot \frac{g_{m11}.R_{12}}{1+g_{m11}.R_{12}} \tag{7}$$

Where;

$g_{m9(11)}$: is the trans-conductance of $M_{9(11)}$.
$R_{10(12)}$: is the output resistor of $M_{10(12)}$.

The proposed cell should be designed well to perform as an ideal buffer with high wide band range. The voltage gain of such cell is equal to unity. To verify that, it is necessary to select proper length of the transistor of M_9, M_{10}, M_{11}, and M_{12}. Equation (7) is derived from the ac model circuit shown in Fig. 5 with the output resistors of M_9 and M_{11} are ignored as they are very large.

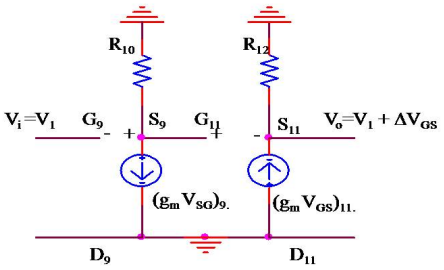


Fig. 5. Ac circuit model for the proposed cell.

III. SIMULATION RESULTS

The complete schematic of the proposed input offset cancellation circuit with DISO amplifier is shown in Fig. 6 being embodiment of the presented block in Fig. 2.

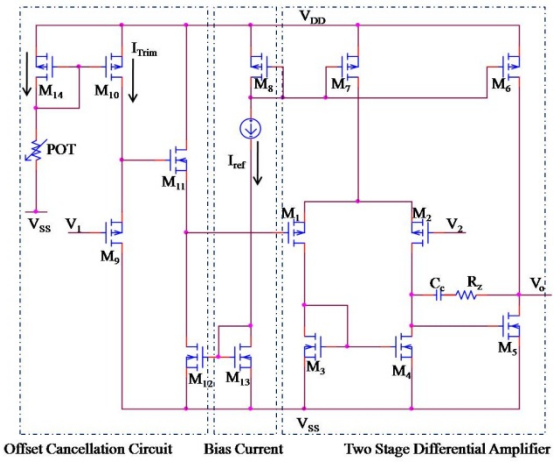


Fig. 6. Schematic of the input offset cancellation with DISO amplifier.

The proposed DISO amplifier having a DC gain of 78 dB, GBW of 58 MHz associated with a phase margin of 67° is designed with the input offset cancellation circuit using the methodology design reported in [6,14]. Table I indicates the dimensions of every transistor and values of C_c and R_z.

The proposed circuit is simulated using spectre to ensure if DISO with input offset cancellation circuit is stable or not. AC response analysis is used to plot the open loop gain to measure the phase margin as shown in Fig. 7. It is worth to mention that the total power consumption of the proposed circuit is less than 0.72mW with I_{ref} equals to 20µA.

TABLE I
SUMMARY OF THE TRANSISTOR'S DIMENSION FOR L=1 µm

Transistor	W(µm)
M_1, M_2	90
M_3,M_4	23
M_5	92
M_6	600
M_7	300
M_8	40
M_9, M_{10}, M_{13},M_{14}	10
M_{11}, M_{12}	23
Cc=2pf,	R_z=1.5kΩ

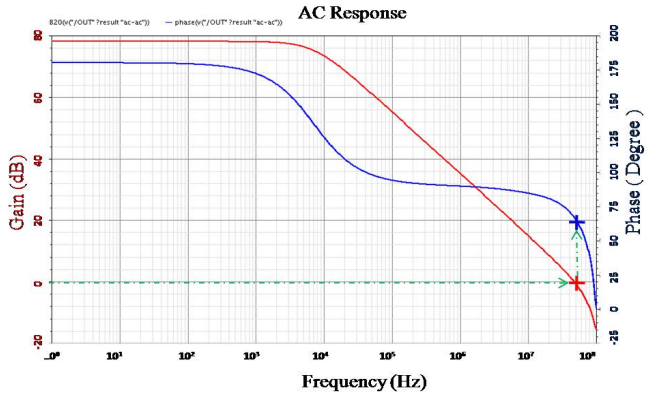


Fig. 7. Loop gain and phase versus frequency.

Figure 8 shows a novel view to sense $V_{os,\,out}$ as a function of current using a Unit Under Test (UUT), normalized resistor (R_n=1Ω).

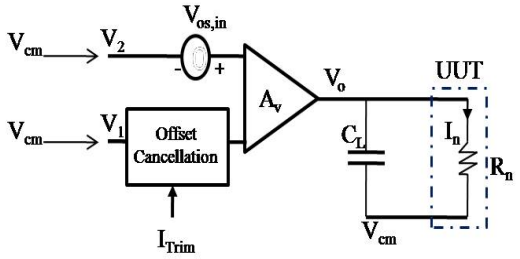


Fig. 8. Sensing circuit for $V_{os,\,out}$ with R_n.

$$V_o = V_{cm} + I_n . R_n \quad (8)$$

The normalized current (I_n) is generated from the difference between currents of M_5 and M_6 and it finds its path through R_n. According to (8), I_n should be equal to zero to revoke any DC error at the output. The injected current, trimming current, is swept to null I_n as shown in Fig. 9. The value of injected current at zero I_n is the optimum current ($I_{optimum}$) to revoke the offset.

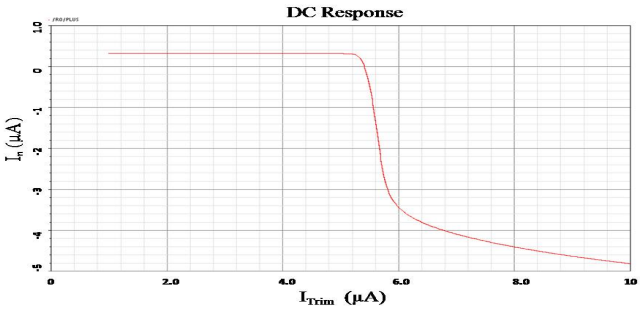


Fig. 9. I_{Trim} versus I_n.

The effect of I_{Trim} on the offset voltage is measured by using a unity feedback test as shown in Fig. 10. V_i is adjusted to be equal 0.75v (mid the range of power supply) and observe the output voltage to be 0.75v. The adjusted value is selected to achieve high dynamic range for input and output. The proposed circuit provides a solution to revoke the offset voltage at $I_{optimum}$ as show in characteristic curve of DISO amplifier in Fig. 10. The relation between $I_{optimum}$ and input DC common voltage (V_{ic}) shown in Fig. 11 is used to set the injected current to be equal $I_{optimum}$ at any desired V_{ic}.

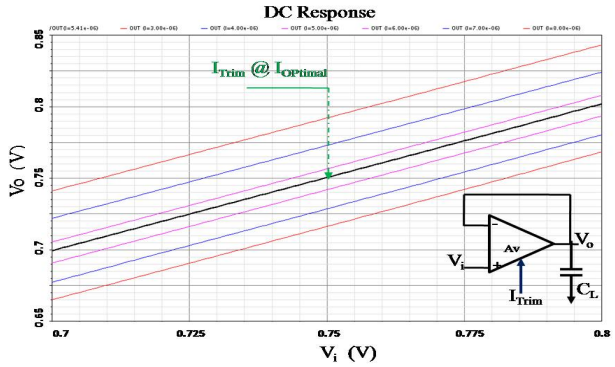


Fig. 10. The DC characteristic curve of DISO with input offset cancellation.

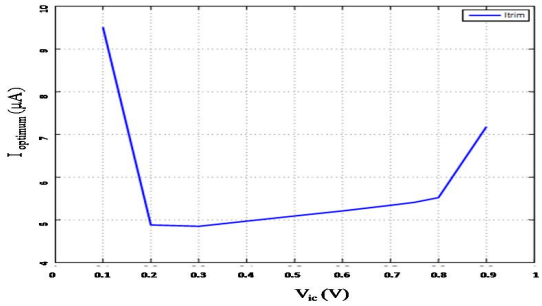


Fig. 11. Common input voltage versus $I_{optimum}$.

Intensive simulation using Monte Carlo tool with models provided by foundry is performed to anticipate any DC error due to mismatches. Monte Carlo tool runs with practical

different correlation coefficients to produce the variation at the output based on statistical analysis. The histogram of the offset is shown in Fig. 12 at V_{ic} equals 0.75v and $I_{optimum}$ =5.41 µA, which confirms the functionality of the proposed circuit solution including significant reduction of the mean offset (Mu) as well as its spread (Sd). Also, Table II indicates statistical result at trimming current equal to $I_{optimum}$ and $I_{non\text{-}optimum}$. The performance of the proposed technique compared to previous work is shown in table III.

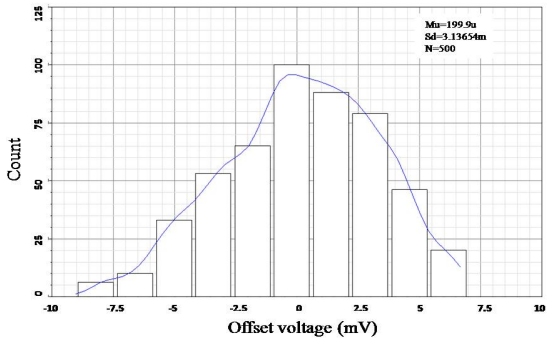


Fig. 12. Histogram simulation of the offset voltage.

TABLE II
STATISTICAL RESULT AT VIC=0.75V

State	Count	Mu	Sd
$I_{Trim}=I_{optimum}$ =5.41 µA	500	199.9µv	3.136mv
$I_{Trim}=I_{non\text{-}optimum}$ = 8µA	500	-33.86mv	3.16mv

TABLE III
THE PERFORMANCE OF THE PROPOSED CIRCUIT AT VIC=0.75V

Item	This work	Ref[18]	Ref[11]	Ref[7]	Ref[19]
Operation	Analog	Digital	Digital	Analog	Analog
Technology	0.13µm CMOS	0.18 µm CMOS	0.35 µm CMOS	CMOS	BJT
Sd	3.13mv	15:3.3mv	-	-	-
Mu	200 µv	-	1.898mv	5m	0.8mv
Power supply	1.5 v	1.8v	3.3v	5v	10v
Power dissipation	0.72mw	10mw	0.6mw	3.25mw	500mw

IV. CONCLUSION

The novel input offset cancellation technique has been presented to provide an analog control line for operational amplifier. This control line produces an injected, trimming, current varied by potentiometer to cancel out DC error. The designed input offset cancellation technique is considered as post-package trim technique in which random and systematic offset have been cancelled out. The proposed input offset cancelation circuit has been used with DISO amplifier, and yields a DC gain of 78 dB, a unity-gain frequency (GBW) of 58 MHz associated with a phase margin of 67°. Moreover, the designed circuit has been simulated and analyzed using statistical and Monte Carlo tool provided from foundry that yields a mean offset 200µv with spread 3.13mv which is a good results according to mentioned former works. Also, the total power consumption is 0.72mW. IBM 0.13µm CMOS technology parameter has been used for circuit simulation with single power supply 1.5v

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